EART11200: Natural Scientist's Toolkit





This week we will look at the burning of fossil fuels, and its consequences. First, consider the ideal gas law

The ideal gas law

The atmosphere is made up of a mixture of gases.

An ideal gas obeys the law

$$PV = NR_{gas}T \quad (PV/T = \text{constant})$$

where P is pressure, V is volume of the gas, T is temperature (in Kelvin), N is number of moles of the gas, and R_{gas} is the universal gas constant (8.314 $JK^{-1}mol^{-1}$)

Throughout most of the atmosphere we can assume the ideal gas law is obeyed.

Be careful with units!

Look back at week 1 if you need to remind yourself about moles.

Pressure units: atmospheric pressure is expressed in SI units of hPa (hecto-Pascal), which is equivalent to the mbar (millibar). Convert to Pascals in calculations (see week 1).

Temperatures must always be expressed in Kelvin for calculations

Comparison of Kelvin, Celsius and Fahrenheit temperature scales

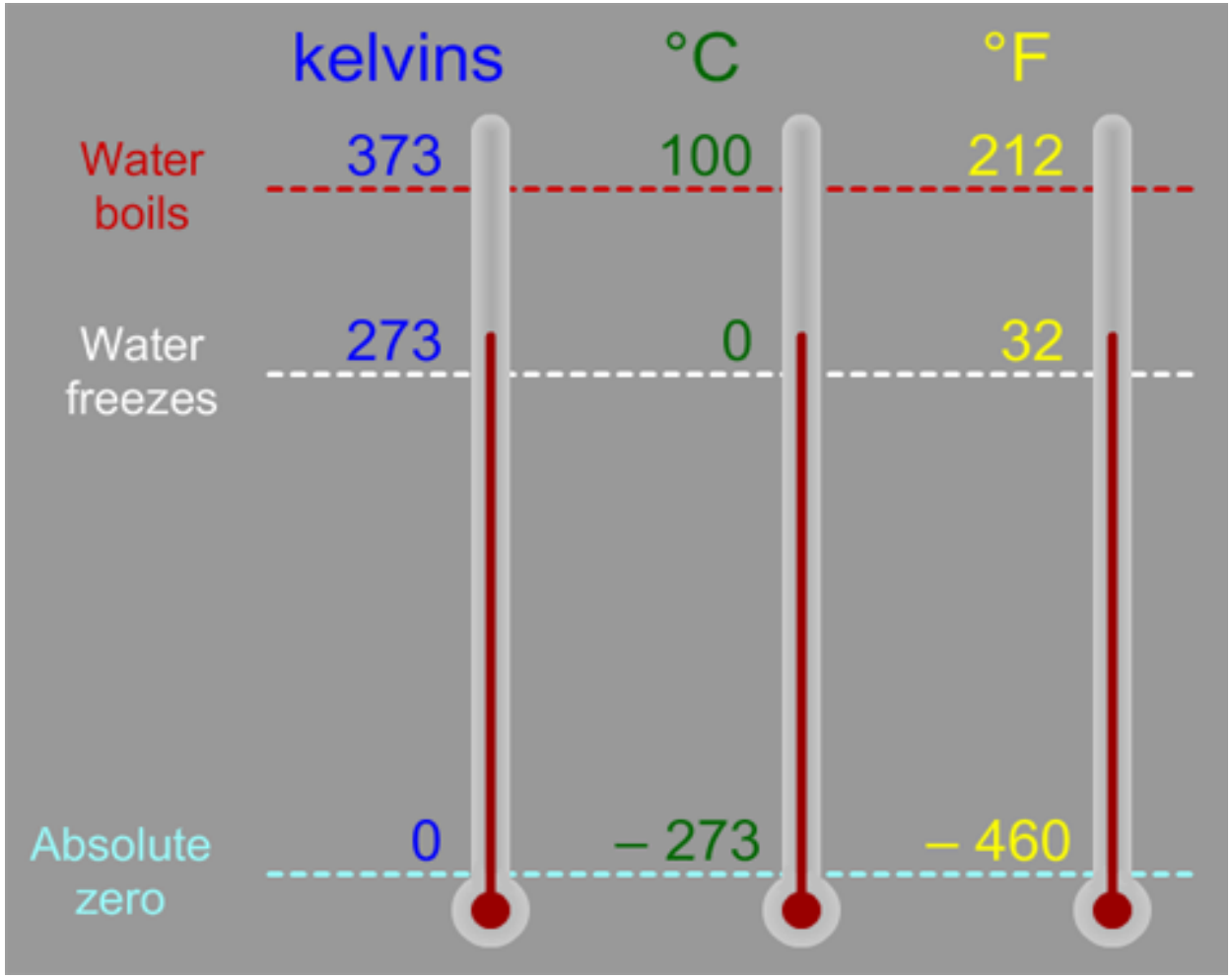
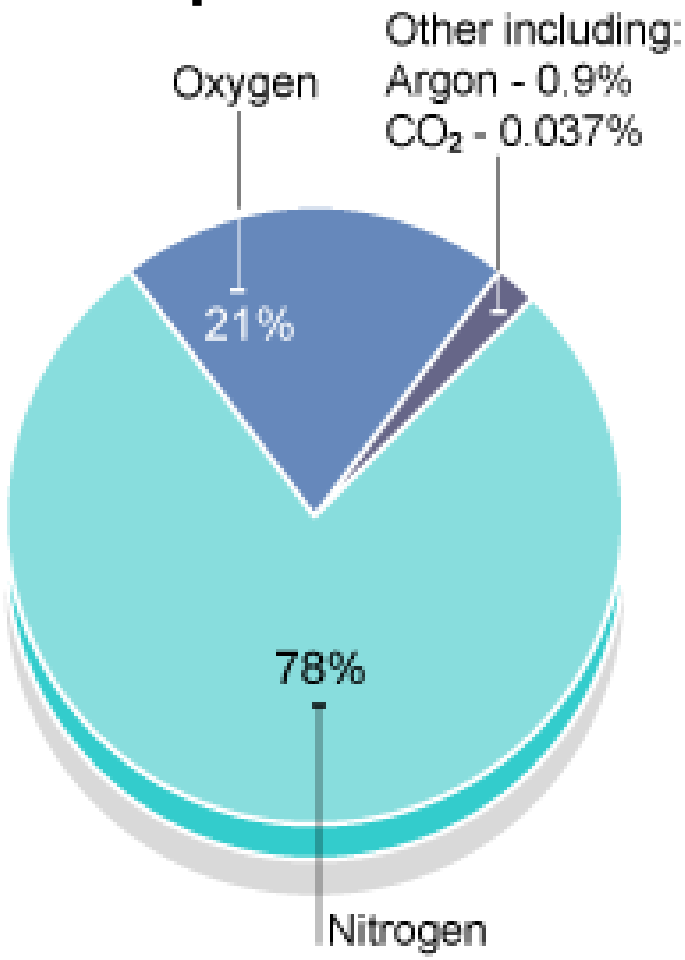


Image courtesy of Windows to the Universe (Randy Russell)

Composition of atmosphere by moles



The atmosphere is a mixture of gases.

Each gas exerts a partial pressure p_i

The ideal gas law holds for each gas:

$p_iV=N_iRT$

The total pressure is the sum of partial pressures:

$p=p_1+p_2+p_3+\ldots+p_N$

We can expand on this idea to reach the concept of ppmv (parts per million by volume), often used to express the concentration of pollutants in the atmosphere.

R=8.314 J K^{-1} mol^{-1}

T=temperature

P=pressure

V=volume

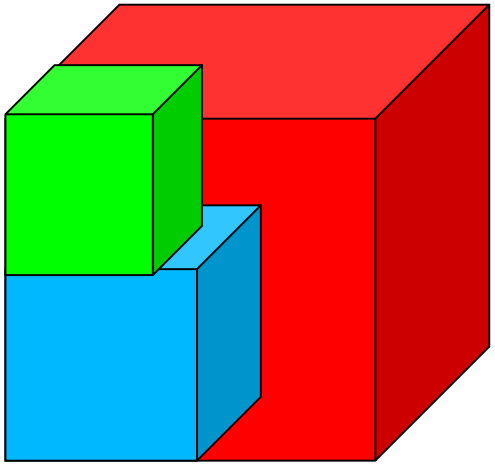
Dalton's law of partial pressures

This states that the total pressure exerted by a mixture of gases is equal to the sum of the pressures exerted by each individual gas.

$$p = p_1 + p_2 + p_3 + \ldots + p_N = \Sigma p_i$$

We can also imagine that all the gases in a box with total pressure P, volume V and temperature T separate themselves out so each gas occupies its own share of the volume (this would not really happen)

Different gases occupy their own volume V_i within the overall volume V, all at the same total pressure P and temperature T



For each gas in its partial volume, the ideal gas law is $PV_i = N_i R_{gas} T$

N_i is the number of moles of the individual gas

For the whole gas $N = N_1 + N_2 + \cdots + N_n$

and also $V = V_1 + V_2 + \cdots + V_n$

The volume occupied by each gas is proportional to the number of moles of each gas (when all at same P and T).

We define the volumetric mixing ratio as

$$\frac{V_i}{V} = \frac{N_i}{N}$$ (so also the molar ratio)

Convention says we use the volumetric mixing ratio, and (because it is usually a small number) multiply by 10E6 to get parts per million by volume (ppmv)